1 Similarity

Last time we saw that if $T \in \mathcal{L}(V)$ for a finite-dimensional vector space V and B, B' are ordered bases of V, then the matrices of T with respect to these bases are related by

$$[T]_{B'} = C^{-1}[T]_B C,$$

where C is the transition matrix from B to B'.



Definition 1.1 (Similar matrices)

Two $n \times n$ matrices A and B are said to be *similar* if there exists an invertible matrix C such that $B = C^{-1}AC$.



Exercise 1.2

Prove that similarity is an equivalence relation on $M^{n\times n}(F)$.



Proof:

Every matrix is similar to itself using the identity matrix, showing reflexivity. If $B = CAC^{-1}$, then $A = C^{-1}BC = (C^{-1})^{-1}B(C^{-1})^{-1}$, and since C is invertible, C^{-1} is invertible, showing symmetry. Finally, if $A = CBC^{-1}$ and $B = EDE^{-1}$, then $A = CEDE^{-1}C^{-1} = (CE)D(CE)^{-1}$, showing transititivty (since products of invertible matrices are again invertible).





Definition 1.3 (Similarity invariants)

A property of a $n \times n$ matrix A which holds for all matrices similar to A is called a *similarity invariant*.



Exercise 1.4

Show that the nullity and rank of a square matrix are similarity invariants. (*Hint:* If $B = C^{-1}AC$ and $\{v_1, ..., v_k\}$ is a basis of Nul$A$, show that $\{C^{-1}v_1, ..., C^{-1}v_k\}$ is a basis of Nul$B$.)



Proof:

Pass to the linear maps perspective. C, C^{-1} are isomorphisms and therefore can be realized as maps $C, C^{-1} : W \to W$ with $B : V \to \text{Im}(B) = W$, $C : W \to W$ and $A : V \to W$. Then, we have that $A = CBC^{-1}$ where C, C^{-1} are isomorphism of W (the image of B). Since these are isomorphisms, these preserve the dimension of the image of B (the rank of B) and preserve the dimension of the kernel of W (the nullity of B).





Remark 1.5

If $B = C^{-1}AC$, then $B^T = C^TA^T(C^{-1})^T = C^TA^T(C^{-1})^T = C^TA^T(C^T)^{-1} = ((C^T)^{-1})^{-1}A^T(C^T)^{-1}$, which shows B^T and A^T are similar. It follows from the exercise that B^T and A^T have the same rank and nullity. Thus, if A and B are similar, then each fundamental matrix space of A has the same dimension as the corresponding fundamental matrix space of B.



Exercise 1.6

Prove that the matrices $A = \begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$.



Solution

Row-reducing, we have:

$$A \to \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix},$$

$$B \to \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

so A is of rank 1 and B is of rank 2, so they cannot be similar.